

kohlhase-omdoc: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00001	10	■ 1.000	\Omega	Ω	tably the Ω MEGA group at Saarland University, the MAYA and ACTIVEMATH
kohlhase-omdoc_F00002	19	■ 0.928	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	distinguish them from presentation markup schemes, and discuss $\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$
kohlhase-omdoc_F00003	19	■ 1.000	$\mathrm{T}_{\mathrm{E}}\mathrm{X}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X}$ is a typesetting markup language that uses explicit markup codes
kohlhase-omdoc_F00004	19	—	$\sqrt{\sin x}$	(not rendered)	—
kohlhase-omdoc_F00005	19	■ 0.878	$\sqrt{\sin x}$	$\sqrt{\sin x}$	$\sqrt{\sin x}$ stands for the mathematical expression $\sqrt{\sin x}$ in $\mathrm{T}_{\mathrm{E}}\mathrm{X}$. To
kohlhase-omdoc_F00006	21	■ 0.696	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	to $\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ in spirit, but tries to give the markup scheme a more declar-
kohlhase-omdoc_F00007	23	■ 0.940	$\langle\!\langle nsURI \rangle\!\rangle$	$\langle\!\langle nsURI \rangle\!\rangle$	descendants are in the namespace $\langle\!\langle nsURI \rangle\!\rangle$, unless they have a namespace
kohlhase-omdoc_F00008	23	■ 0.198	$\langle$	$\langle$	by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\langle nsURI \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is
kohlhase-omdoc_F00009	23	■ 0.198	$\rangle=$	$\rangle=$	by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\langle nsURI \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is
kohlhase-omdoc_F00010	23	■ 0.198	$\rangle$	$\rangle$	by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\langle nsURI \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is
kohlhase-omdoc_F00011	23	■ 0.198	$\langle\!\langle$	$\langle\!\langle$	by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\langle nsURI \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is
kohlhase-omdoc_F00012	23	■ 0.198	$\rangle\!\rangle$	$\rangle\!\rangle$	by a pseudo-attribute of the form $\mathrm{xmlns}:\langle\!\langle nsURI \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is
kohlhase-omdoc_F00013	23	■ 0.190	$\langle\!\langle nsURI \rangle\!\rangle$	$\langle\!\langle nsURI \rangle\!\rangle$	a qualified name : a pair $\langle\!\langle nsURI \rangle\!\rangle:\langle\!\langle el \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is a namespace abbrevi-
kohlhase-omdoc_F00014	23	■ 0.190	$\langle\!\langle nsURI \rangle\!\rangle$	$\langle\!\langle nsURI \rangle\!\rangle$	a qualified name : a pair $\langle\!\langle nsURI \rangle\!\rangle:\langle\!\langle el \rangle\!\rangle$, where $\langle\!\langle nsURI \rangle\!\rangle$ is a namespace abbrevi-
kohlhase-omdoc_F00015	23	■ 0.917	$\hat{=}$	$\hat{=}$	$\hat{=}$
kohlhase-omdoc_F00016	25	■ 0.832	7^{th}	7^{th}	the 7^{th} child of the 3^{rd} child of the (unique) element that has ID-type attribute
kohlhase-omdoc_F00017	25	■ 0.832	3^{rd}	3^{rd}	the 7^{th} child of the 3^{rd} child of the (unique) element that has ID-type attribute
kohlhase-omdoc_F00018	25	■ 0.984	D	D	(the shorthand xpointer): If $\langle\!\langle uri \rangle\!\rangle$ is a URI of an XML document D then
kohlhase-omdoc_F00019	25	■ 0.454	$\langle\!\langle id \rangle\!\rangle$	$\langle\!\langle id \rangle\!\rangle$	ID with value $\langle\!\langle id \rangle\!\rangle$.
kohlhase-omdoc_F00020	27	■ 0.990	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$	systems like $\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$.
kohlhase-omdoc_F00021	28	■ 0.348	$\{ \}^{\{ + \}}$	$\{ \}^{\{ + \}}$	formulae for the Web are MATHML [ABC ⁺ 03] and OPENMATH [BCC ⁺ 04].
kohlhase-omdoc_F00022	28	■ 1.000	$\{ \}^{\{ \text{®} \}}$	®	mathematical software packages like Wolfram Research’s MATHEMATICA [®]
kohlhase-omdoc_F00023	29	■ 1.000	$\frac{3}{x+2}$	$\frac{3}{x+2}$	the expression $\frac{3}{x+2}$ would be represented as the layout tree in Figure 2.2.

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kohlhase-omdoc_ F00024	29	■ 0.991	\Sigma	Σ	DTD, e.g. <code>&sum</code> ; for Σ .
kohlhase-omdoc_ F00025	29	■ 0.840	14{ }^{\text{th}}	14 th	fixed set of mathematical concepts – the K-14 fragment (Kindergarten to 14 th
kohlhase-omdoc_ F00026	30	■ 0.998	\mathrm{T}_{\mathrm{E}} \mathrm{X}^3	$\mathrm{T}_\mathrm{E}X^3$	of $\mathrm{T}_\mathrm{E}X$. However, it does not offer the programming language features of $\mathrm{T}_\mathrm{E}X^3$
kohlhase-omdoc_ F00027	30	■ 1.000	{ }^3 \mathrm{~T}_{\mathrm{E}} \mathrm{X}	${}^3 \mathrm{T}_\mathrm{E}X$	${}^3 \mathrm{T}_\mathrm{E}X$ contains a full, Turing-complete – if somewhat awkward – programming lan-
kohlhase-omdoc_ F00028	33	■ 1.000	\forall a, b . a+b=b+a	$\forall a, b. a + b = b + a$	(the formula $\forall a, b. a + b = b + a$) in OPENMATH. Note that as we have dis-
kohlhase-omdoc_ F00029	35	■ 1.000	S=(G, \circ)	$S = (G, \circ)$	A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$, such that
kohlhase-omdoc_ F00030	35	■ 1.000	e \in G	$e \in G$	A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$, such that
kohlhase-omdoc_ F00031	35	■ 0.988	e \circ x=x	$e \circ x = x$	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .
kohlhase-omdoc_ F00032	35	■ 0.988	x \in G	$x \in G$	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .
kohlhase-omdoc_ F00033	35	■ 0.988	e	e	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .
kohlhase-omdoc_ F00034	35	■ 0.988	S	S	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .
kohlhase-omdoc_ F00035	35	■ 0.915	f	f	Proof: We assume that there is another left unit $f \dots$
kohlhase-omdoc_ F00036	35	—	\square	$\square$	$\square$
kohlhase-omdoc_ F00037	37	■ 0.491	\mathrm{BCF}^+	BCF^+	MIZAR [Rud92] and Ω MEGA [BCF ⁺ 97] which utilize strong logical systems
kohlhase-omdoc_ F00038	38	■ 0.995	\left[\mathrm{SBB}^+{}_{02}\right]	[SBB ⁺ 02]	on the other. Ω MEGA [SBB ⁺ 02] is a large-scale proof development environ-
kohlhase-omdoc_ F00039	40	■ 0.772	1{ }^{\text{st}} 2000	1 st 2000	Version 1.0 of the OMDoc format was released on November 1 st 2000 to
kohlhase-omdoc_ F00040	40	■ 0.981	29{ }^{\text{th}} 2001	29 th 2001	OMDoc 1.1 was released on December 29 th 2001 as an attempt to roll
kohlhase-omdoc_ F00041	41	■ 0.996	X+0=0</ \mathrm{FMP}>	$X + 0 = 0 < / \mathrm{FMP} >$	$< \mathrm{FMP} > X + 0 = 0 < / \mathrm{FMP} >$
kohlhase-omdoc_ F00042	41	■ 0.999	<\mathrm{FMP}>X+s(Y)=s(X+Y)</ \mathrm{FMP}>	$< \mathrm{FMP} > X + s(Y) = s(X + Y) < / \mathrm{FMP} >$	$< \mathrm{FMP} > X + s(Y) = s(X + Y) < / \mathrm{FMP} >$
kohlhase-omdoc_ F00043	42	■ 0.712	{ }^+{}_{04}	⁺ 04	mulae, OMDoc uses the established standards OPENMATH [BCC ⁺ 04]
kohlhase-omdoc_ F00044	43	■ 0.501	{ }^{\text{TM}}	TM	On the left hand side we see CAS like MATHEMATICA [®] [Wol02] or MAPLE [™]
kohlhase-omdoc_ F00045	43	■ 0.820	\left[\mathrm{CGG}^+{}_{92}\right]	[CGG ⁺ 92]	[CGG ⁺ 92] that are relatively restricted in the mathematical objects — they
kohlhase-omdoc_ F00046	48	■ 0.988	\mathrm{LAT}_{\mathrm{E}} \mathrm{X}	$\mathrm{LAT}_\mathrm{E}X$	the $\mathrm{T}_\mathrm{E}X/\mathrm{L}\mathrm{A}\mathrm{T}_\mathrm{E}X$ system, we will take a $\mathrm{L}\mathrm{A}\mathrm{T}_\mathrm{E}X$ representation as the starting
kohlhase-omdoc_ F00047	49	■ 0.851	E	E	DEFINITION 1. Let E be a set. A mapping of $E \times E$ is called a law of composition
kohlhase-omdoc_ F00048	49	■ 0.851	E \times E	$E \times E$	DEFINITION 1. Let E be a set. A mapping of $E \times E$ is called a law of composition
kohlhase-omdoc_ F00049	49	■ 0.988	f(x, y)	$f(x, y)$	on E . The value $f(x, y)$ of f for an ordered pair $(x, y) \in E \times E$ is called the
kohlhase-omdoc_ F00050	49	■ 0.988	(x, y) \in E \times E	$(x, y) \in E \times E$	on E . The value $f(x, y)$ of f for an ordered pair $(x, y) \in E \times E$ is called the
kohlhase-omdoc_ F00051	49	■ 0.965	x	x	composition of x and y under this law. A set with a law of composition is called

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kohlhase-omdoc_F00052	49	■ 0.965	y	y	composition of x and y under this law. A set with a law of composition is called
kohlhase-omdoc_F00053	49	■ 0.999	$x+y, x \cdot y$	$x + y, x.y$	symbols the composition of x and y is written respectively as $x + y, x.y$ or xy .
kohlhase-omdoc_F00054	49	■ 0.999	$x \ y$	xy	symbols the composition of x and y is written respectively as $x + y, x.y$ or xy .
kohlhase-omdoc_F00055	49	■ 1.000	$x+y$	$x + y$	$x + y$ being called the <i>sum</i> of x and y) and we say that it is <i>written additively</i> ;
kohlhase-omdoc_F00056	49	■ 0.999	$x \cdot y = x \ y$	$x.y = xy$	<u>$x.y = xy$</u> being called the <i>product</i> for x and y) and we say that it is <i>written</i>
kohlhase-omdoc_F00057	49	■ 0.964	$\top$	$\top$	use the symbols $\top$ and $\perp$ to denote arbitrary laws of composition.
kohlhase-omdoc_F00058	49	■ 1.000	$(X, Y) \mapsto X \cup Y$	$(X, Y) \mapsto X \cup Y$	<i>Examples.</i> (1) The mappings $(X, Y) \mapsto X \cup Y$ and $(X, Y) \mapsto X \cap Y$ are
kohlhase-omdoc_F00059	49	■ 1.000	$(X, Y) \mapsto X \cap Y$	$(X, Y) \mapsto X \cap Y$	<i>Examples.</i> (1) The mappings $(X, Y) \mapsto X \cup Y$ and $(X, Y) \mapsto X \cap Y$ are
kohlhase-omdoc_F00060	49	■ 1.000	$x \in \mathbf{N}$	$x \in \mathbf{N}$	mentation are laws of composition (the compositions of $x \in \mathbf{N}$ and $y \in \mathbf{N}$
kohlhase-omdoc_F00061	49	■ 1.000	$y \in \mathbf{N}$	$y \in \mathbf{N}$	mentation are laws of composition (the compositions of $x \in \mathbf{N}$ and $y \in \mathbf{N}$
kohlhase-omdoc_F00062	49	■ 0.997	$x+y, x \ y$	$x + y, xy$	under these laws being denoted respectively by $x + y, xy$, or $x.y$ and x^y) (<i>Set</i>
kohlhase-omdoc_F00063	49	■ 0.997	$x \cdot y$	$x.y$	under these laws being denoted respectively by $x + y, xy$, or $x.y$ and x^y) (<i>Set</i>
kohlhase-omdoc_F00064	49	■ 0.997	$x^{\{y\}}$	x^y	under these laws being denoted respectively by $x + y, xy$, or $x.y$ and x^y) (<i>Set</i>
kohlhase-omdoc_F00065	49	■ 1.000	$(X, Y) \mapsto X \circ Y$	$(X, Y) \mapsto X \circ Y$	(3) Let E be a set; the mapping <u>$(X, Y) \mapsto X \circ Y$</u> is a law of composition
kohlhase-omdoc_F00066	49	■ 1.000	$(f, g) \mapsto f \circ g$	$(f, g) \mapsto f \circ g$	mapping <u>$(f, g) \mapsto f \circ g$</u> is a law of composition on the set of mappings from E
kohlhase-omdoc_F00067	52	■ 0.221	$\mathrm{APT}_{\mathrm{E}}\mathrm{X}$	$\mathrm{APT}_{\mathrm{E}}\mathrm{X}$	The text of the L ^A T _E X fragment we are migrating. For simplic-
kohlhase-omdoc_F00068	59	■ 0.999	λ	λ	a λ -term $\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$. The value simple of the type at-
kohlhase-omdoc_F00069	59	■ 0.999	$\lambda E, F. \mathit{operatorname{set}}(E) \wedge F : E \times E \rightarrow E^2$	$\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$	a λ -term <u>$\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$</u> . The value simple of the type at-
kohlhase-omdoc_F00070	59	■ 1.000	(A, B)	(A, B)	it occurs. So if we have <code>law_of_composition(A, B)</code> somewhere this can be
kohlhase-omdoc_F00071	59	■ 0.863	$(\lambda E, F. \mathit{operatorname{set}}(E) \wedge F : E \times E \rightarrow E)(A, B)$	$(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$	reduced to <u>$(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$</u> which in turn reduces ³
kohlhase-omdoc_F00072	59	■ 0.917	$\mathit{operatorname{set}}(A) \wedge B : A \times A \rightarrow A$	$\mathit{set}(A) \wedge B : A \times A \rightarrow A$	to <u>$\mathit{set}(A) \wedge B : A \times A \rightarrow A$</u> or in other words <code>law_of_composition(A, B)</code> is
kohlhase-omdoc_F00073	59	■ 1.000	A	A	true, iff A is a set and B is a function from $A \times A$ to A . This definition is
kohlhase-omdoc_F00074	59	■ 1.000	B	B	true, iff A is a set and B is a function from $A \times A$ to A . This definition is
kohlhase-omdoc_F00075	59	■ 1.000	$A \times A$	$A \times A$	true, iff A is a set and B is a function from $A \times A$ to A . This definition is
kohlhase-omdoc_F00076	59	■ 0.819	$\lambda X. A$	$\lambda X. A$	the form $\lambda X. A$ to an argument B , then the result is obtained by binding all the
kohlhase-omdoc_F00077	59	■ 1.000	X	X	formal parameters X to the actual parameter B , i.e. the result is the value of A ,
kohlhase-omdoc_F00078	60	■ 0.967	M	M	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of
kohlhase-omdoc_F00079	60	■ 0.967	(E, F)	(E, F)	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of

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kohlhase-omdoc_ F00080	60	■ 0.967	F	F	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of
kohlhase-omdoc_ F00081	62	■ 0.986	$\left[\mathrm{BCC}^{+04}\right]$	$\left[\mathrm{BCC}^{+04}\right]$	dard $\left[\mathrm{BCC}^{+04}\right]$ to codify knowledge about mathematical symbols and con-
kohlhase-omdoc_ F00082	66	■ 1.000	n	n	The symbol to represent the n -ary function to return the least common mul-
kohlhase-omdoc_ F00083	66	■ 0.981	$\operatorname{lcm}(a, b)$	$\operatorname{lcm}(a, b)$	We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$
kohlhase-omdoc_ F00084	66	■ 0.981	$\frac{a \cdot b}{\operatorname{gcd}(a, b)}$	$\frac{a \cdot b}{\operatorname{gcd}(a, b)}$	We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$
kohlhase-omdoc_ F00085	66	■ 0.940	a	a	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_ F00086	66	■ 0.940	b	b	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_ F00087	66	■ 0.940	$c > 0$	$c > 0$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_ F00088	66	■ 0.940	$(a \mid c)$	$(a \mid c)$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_ F00089	66	■ 0.940	$(b \mid c)$	$(b \mid c)$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_ F00090	66	■ 1.000	$c < \operatorname{lcm}(a, b)$	$c < \operatorname{lcm}(a, b)$	$c < \operatorname{lcm}(a, b)$.
kohlhase-omdoc_ F00091	66	■ 0.998	$\operatorname{kgV}(a, b)$	$\operatorname{kgV}(a, b)$	Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$
kohlhase-omdoc_ F00092	66	■ 0.998	$\frac{a \cdot b}{\operatorname{ggT}(a, b)}$	$\frac{a \cdot b}{\operatorname{ggT}(a, b)}$	Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$
kohlhase-omdoc_ F00093	66	■ 0.354	$b \mid c$	$b \mid c$	Lemma 3. Für alle ganzen Zahlen a, b gibt es kein $c > 0$ mit $(a c)$ und $(b c)$
kohlhase-omdoc_ F00094	66	■ 0.531	$c < \operatorname{kgV}(a, b)$	$c < \operatorname{kgV}(a, b)$	und $c < \operatorname{kgV}(a, b)$.
kohlhase-omdoc_ F00095	68	■ 1.000	$\mathbb{N}$	$\mathbb{N}$	more elaborate definition of $\mathbb{N}$). The <code>sortdef</code> element posits that the set of
kohlhase-omdoc_ F00097	68	■ 1.000	s	s	instance the number three is represented as $s(s(s(0)))$, where s stands for
kohlhase-omdoc_ F00098	68	■ 1.000	$\leq$	$\leq$	Finally we have extended the natural numbers by an ordering function $\leq$
kohlhase-omdoc_ F00099	69	■ 0.887	$\mathrm{T0}^1$	$\mathrm{T0}^1$	theory, as witnessed by the assertion $\operatorname{leq}.\mathrm{T0}^1$.
kohlhase-omdoc_ F00100	72	■ 1.000	$\sigma(\circ)$	$\sigma(\circ)$	together with the converse $\sigma(\circ)$ of $\circ$ is also a semigroup: the obligation for
kohlhase-omdoc_ F00101	72	■ 1.000	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$, which is logically equivalent
kohlhase-omdoc_ F00102	72	■ 0.835	τx	τx	scription operator τx .: If there is a unique x , such that A is true, then the
kohlhase-omdoc_ F00103	73	■ 1.000	ρ	ρ	ρ in Figure 7.1), in OMDoc we can share this structure using the <code>base on</code>
kohlhase-omdoc_ F00104	73	■ 0.951	$\cdot^{-1}$	$\cdot^{-1}$	function $\cdot^{-1}$ from that via a definite description.
kohlhase-omdoc_ F00105	74	■ 1.000	σ	σ	represented by σ and τ in Figure 7.1). These rename all of the imported
kohlhase-omdoc_ F00106	74	■ 1.000	τ	τ	represented by σ and τ in Figure 7.1). These rename all of the imported
kohlhase-omdoc_ F00107	74	■ 0.301	$\cdot^*$	$\cdot^*$	the punctuated set constructor $\cdot^*$, which is imported from the additive group
kohlhase-omdoc_ F00108	74	■ 0.267	l	l	$\langle \operatorname{morphism} \rangle M \mapsto R, x \circ y \mapsto x * y, e \mapsto 1, \cdot^{-1} \mapsto - \langle / \operatorname{morphism} \rangle$

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kohlhase-omdoc_F00109	84	■ 1.000	$\mathcal{N}$	$\mathcal{N}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00110	84	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00111	84	■ 1.000	$\mathcal{E}_{\mathcal{N}}$	$\mathcal{E}_{\mathcal{N}}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00112	84	■ 1.000	$\mathcal{S}$	$\mathcal{S}$	is insufficient. Then we propose a first (straw-man) solution $\mathcal{S}$, and show an
kohlhase-omdoc_F00113	84	■ 1.000	$\mathcal{E}_{\mathcal{S}}$	$\mathcal{E}_{\mathcal{S}}$	example $\mathcal{E}_{\mathcal{S}}$ of why this does not work. Based on the information we gleaned
kohlhase-omdoc_F00114	84	■ 1.000	$\mathcal{E}_{\mathcal{F}}$	$\mathcal{E}_{\mathcal{F}}$	theory and demonstrate that this works on $\mathcal{E}_{\mathcal{F}}$.
kohlhase-omdoc_F00115	84	■ 1.000	$\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$	$\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$	sents the content structure, where the boxes $\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$, and $\mathcal{E}_{\mathcal{F}}$ signify
kohlhase-omdoc_F00116	84	■ 0.384	$\operatorname{slides} \mathrm{sl}_i$	slides sl_i	the slides sl_i are grouped into a lecture. The dashed lines between the two
kohlhase-omdoc_F00117	84	■ 1.000	n_1	n_1	example: the text fragment n_1 links the content $\mathcal{E}_{\mathcal{N}}$, which is referenced from
kohlhase-omdoc_F00118	84	■ 0.974	n_2	n_2	the content structure, to slide 1. The fragment n_2 might say something like
kohlhase-omdoc_F00119	85	■ 0.986	$\left[\operatorname{MBG}^{+03}\right]$	$\left[\operatorname{MBG}^{+03}\right]$	the ACTIVEMATH system $\left[\operatorname{MBG}^{+03}\right]$ generates a simple narrative structure
kohlhase-omdoc_F00120	87	■ 0.810	9.3^2	9.3^2	in Listing 9.3 ² to the URL which uniquely identifies the service object in the
kohlhase-omdoc_F00121	103	■ 1.000	R	R	Let R be a ref element of type <code>include</code> . We call the element the URI
kohlhase-omdoc_F00122	103	■ 0.978	$\left[\operatorname{SBC}^{+00}\right]$	$\left[\operatorname{SBC}^{+00}\right]$	system $\left[\operatorname{SBC}^{+00}\right]$ can — for instance — extract document fragments based on
kohlhase-omdoc_F00123	107	■ 0.910	$\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$	$\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$	Concretely, the format is $\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle T \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$
kohlhase-omdoc_F00124	107	■ 0.781	$\langle YYY Y \rangle$	$\langle YYY Y \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00125	107	■ 0.781	$\langle MM \rangle$	$\langle MM \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00126	107	■ 0.781	$\langle DD \rangle$	$\langle DD \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00127	107	■ 0.582	$\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$	$\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$	date/time separator and $\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$ represent hour, minutes, and
kohlhase-omdoc_F00128	107	■ 0.260	$\langle ss \rangle . \langle sss \rangle$	$\langle ss \rangle . \langle sss \rangle$	cision of fractional seconds if desired, i.e the format $\langle ss \rangle . \langle sss \rangle$ with
kohlhase-omdoc_F00129	108	■ 1.000	$\widehat{=}$	$\widehat{=}$	be an ISO 639 norm two-letter language specifier, like <code>en</code> $\widehat{=}$ English,
kohlhase-omdoc_F00130	108	■ 0.327	$\mathrm{de} \widehat{=}$	$\mathrm{de} \widehat{=}$	<code>de</code> $\widehat{=}$ German, <code>fr</code> $\widehat{=}$ French, <code>nl</code> $\widehat{=}$ Dutch, . . .
kohlhase-omdoc_F00131	114	■ 0.586	$^{+03}$	$^{+03}$	and the MATHML 2.0 recommendation (second edition) $\left[\operatorname{ABC}^{+03}\right]$ are part
kohlhase-omdoc_F00132	116	■ 1.000	$\sin(x)$	$\sin(x)$	For instance, the formula $\sin(x)$ would be modeled as an application of
kohlhase-omdoc_F00133	116	■ 1.000	$\forall x. \sin(x) \leq \pi$	$\forall x. \sin(x) \leq \pi$	formula $\forall x. \sin(x) \leq \pi$.
kohlhase-omdoc_F00134	117	■ 0.999	$x + \pi$	$x + \pi$	In the OPENMATH fragment in Listing 13.3 the expression $x + \pi$ is annotated

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kohlhase-omdoc_ F00135	118	■ 1.000	<code>\sin (\pi)</code>	$\sin(\pi)$	faulty representation of $\sin(\pi)$. The outer error flags an arity violation (the
kohlhase-omdoc_ F00136	118	■ 1.000	<code>\pi</code>	π	the representation of the constant π (we used the name po instead of pi).
kohlhase-omdoc_ F00137	121	■ 0.973	<code>14^{\text{th}}</code>	14 th	matics (Kindergarten to 14 th grade (i.e. undergraduate college level)).
kohlhase-omdoc_ F00138	121	■ 0.630	<code>2.0\left[\mathrm{ABC}^{+}03\right]</code>	2.0 [ABC ⁺ 03]	second edition of MATHML 2.0 [ABC ⁺ 03], a maintenance release for the
kohlhase-omdoc_ F00139	121	■ 0.997	<code>\left[\mathrm{ABC}^{+}03\right]</code>	[ABC ⁺ 03]	MATHML 2.0 that are relevant to OMDOC; for the full story see [ABC⁺03] .
kohlhase-omdoc_ F00140	121	■ 0.456	<code>^{\{7\}}</code>	$\overline{7}$	The top-level element of MATHML is the m:math ⁷ element, see Figure 13.10
kohlhase-omdoc_ F00141	124	■ 1.000	<code>(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})</code>	$(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$	e.g. the type $(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$ for functionals on the real numbers. In
kohlhase-omdoc_ F00142	126	■ 0.397	<code>\mathrm{cd}=</code>	cd =	tent dictionary identified in the cd attribute. A symbol <code><OMS cd="⟨cd₁⟩"</code>
kohlhase-omdoc_ F00143	126	■ 0.397	<code>\left\langle\mathrm{cd}_1\right\rangle</code>	$\langle\langle cd_1 \rangle\rangle$	tent dictionary identified in the cd attribute. A symbol <code><OMS cd="⟨cd₁⟩"</code>
kohlhase-omdoc_ F00144	126	■ 0.175	<code>\left\langle\mathrm{cd}_2\right\rangle^{\prime}</code>	$\langle\langle cd_2 \rangle\rangle'$	<code>name="⟨name₁⟩"/></code> is equal to <code><OMS cd="⟨cd₂⟩" name="⟨name₂⟩"/></code> , iff
kohlhase-omdoc_ F00145	126	■ 0.175	<code>\left\langle\mathrm{cd}_2\right\rangle</code>	$\langle\langle cd_2 \rangle\rangle$	<code>name="⟨name₁⟩"/></code> is equal to <code><OMS cd="⟨cd₂⟩" name="⟨name₂⟩"/></code> , iff
kohlhase-omdoc_ F00146	126	■ 0.973	<code>\left\langle\mathrm{cd}_1\right\rangle=\left\langle\mathrm{cd}_2\right\rangle</code>	$\langle\langle cd_1 \rangle\rangle = \langle\langle cd_2 \rangle\rangle$	$\langle\langle cd_1 \rangle\rangle = \langle\langle cd_2 \rangle\rangle$ and $\langle\langle name_1 \rangle\rangle = \langle\langle name_2 \rangle\rangle$ as XML simple names. In case of vari-
kohlhase-omdoc_ F00147	126	■ 0.973	<code>\left\langle\mathrm{cd}_1\right\rangle=\left\langle\mathrm{cd}_2\right\rangle</code>	$\langle\langle name_1 \rangle\rangle = \langle\langle name_2 \rangle\rangle$	$\langle\langle cd_1 \rangle\rangle = \langle\langle cd_2 \rangle\rangle$ and $\langle\langle name_1 \rangle\rangle = \langle\langle name_2 \rangle\rangle$ as XML simple names. In case of vari-
kohlhase-omdoc_ F00148	126	■ 0.923	<code>\alpha</code>	α	named without changing the object (α-conversion). If <code><OMV name="x"/></code>
kohlhase-omdoc_ F00149	129	■ 0.990	<code>\mathrm{CMP}^2</code>	$\mathbf{CMP}^2$	is represented as a groups of CMP ² elements which contain the vernacular
kohlhase-omdoc_ F00150	130	■ 0.942	<code>\mathrm{FMP}^4</code>	$\mathbf{FMP}^4$	An FMP ⁴ element is the general element for representing formal mathematical
kohlhase-omdoc_ F00151	131	■ 0.287	<code>V, F. Im (F) = V</code>	$V, F. \mathbf{Im}(F) = V$	<code><FMP logic="hol">binop = \lambda V, F. \mathbf{Im}(F) = V \wedge \mathbf{Dom}(F) = V</FMP></code>
kohlhase-omdoc_ F00152	131	■ 0.404	<code>A</ \mathrm{FMP}></code>	$A < / \mathbf{FMP} >$	is equivalent to <code><FMP>A</FMP></code> , where <i>A</i> is an OPENMATH, Content-MATHML,
kohlhase-omdoc_ F00153	134	■ 0.562	<code>0 \in \mathbb{N}</code>	$0 \in \mathbb{N}$	ken ⁵ . Let us consider an example: We can equivalently say “ $0 \in \mathbb{N}$ ” and “the
kohlhase-omdoc_ F00154	137	■ 0.995	<code>\underline{\mathrm{ul}}</code>	$\underline{\mathbf{ul}}$	<i>Unordered Lists</i> ul is the constructor for unordered or bulleted lists, the in
kohlhase-omdoc_ F00155	140	■ 1.000	<code>s(n)</code>	$s(n)$	2. The successor $s(n)$ of a natural number n is a natural number.
kohlhase-omdoc_ F00156	140	■ 0.997	<code>0, s</code>	$0, s$	structures that interpret 0, <i>s</i> , and $\mathbb{N}$ and satisfy these axioms must be isomor-
kohlhase-omdoc_ F00157	141	■ 1.000	<code>\mathcal{T}</code>	$\mathcal{T}$	of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$
kohlhase-omdoc_ F00158	141	■ 1.000	<code>s=</code>	$s =$	of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$
kohlhase-omdoc_ F00159	141	■ 0.623	<code>\mathbf{A}</code>	A	A , where <i>s</i> is a symbol that does not occur in $\mathcal{T}$ and A is a formula containing
kohlhase-omdoc_ F00160	141	■ 1.000	<code>s=\mathbf{A}</code>	$s = \mathbf{A}$	symbols. The axiom $s = \mathbf{A}$ only constrains the meaning of the new symbol
kohlhase-omdoc_ F00161	141	■ 1.000	<code>\mathcal{A}</code>	$\mathcal{A}$	Generally an axiom $\mathcal{A}$ that results in a conservative extension is called

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kohlhase-omdoc_F00163	141	■ 1.000	$e^{\cdot}$	$e^{\cdot}$	$e^{\cdot}$
kohlhase-omdoc_F00164	141	■ 0.554	$\partial f = f[$	$\partial f = f[$	the differential equation $\partial f = f$ [where $f(0) = 1$].
kohlhase-omdoc_F00165	141	■ 0.554	$f(0)=1]$	$f(0) = 1]$	the differential equation $\partial f = f$ [where $f(0) = 1$].
kohlhase-omdoc_F00166	141	■ 1.000	$x+\theta=x$	$x + 0 = x$	the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.
kohlhase-omdoc_F00167	141	■ 1.000	$x+s(y)=s(x+y)$	$x + s(y) = s(x + y)$	the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.
kohlhase-omdoc_F00168	143	■ 0.993	$1.1\left[\mathrm{BPSM}^{+04}\right]$	1.1 [BPSM ⁺ 04]	1.1 [BPSM⁺04] . The optional attribute scope takes the values global and
kohlhase-omdoc_F00169	144	■ 1.000	$e: t$	$e : t$	processes. A type declaration $e: t$ makes the information that a symbol or
kohlhase-omdoc_F00170	145	■ 0.998	t	t	expression e is in a set represented by a type t available to a specified math-
kohlhase-omdoc_F00171	145	■ 1.000	$\sum_{i=1}^n a_i x^i$	$\sum_{i=1}^n a_i x^i$	or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the a_i are real
kohlhase-omdoc_F00172	145	■ 1.000	a_i	a_i	or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the a_i are real
kohlhase-omdoc_F00173	145	■ 0.999	$\forall x_1, \dots, x_n. e \in t$	$\forall x_1, \dots, x_n. e \in t$	$\forall x_1, \dots, x_n. e \in t$ for a type declaration $e: t$ with variables $x_1, \dots, x_n$. If the
kohlhase-omdoc_F00174	145	■ 0.999	$x_1, \dots, x_n$	$x_1, \dots, x_n$	$\forall x_1, \dots, x_n. e \in t$ for a type declaration $e: t$ with variables $x_1, \dots, x_n$. If the
kohlhase-omdoc_F00176	149	■ 0.944	$x+x$	$x + x$	Listing 15.14 contains a type assertion $x+x: evens$, which makes the infor-
kohlhase-omdoc_F00177	150	■ 1.000	$\mathcal{C}_i$	$\mathcal{C}_i$	and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-
kohlhase-omdoc_F00180	150	■ 1.000	$\mathcal{F}^{-1}$	$\mathcal{F}^{-1}$	and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-
kohlhase-omdoc_F00181	308	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	(i.e. a set of well-formed formulae) and a calculus $\mathcal{C}$ (i.e. a set of inference
kohlhase-omdoc_F00186	150	■ 0.999	$\mathcal{A}_1, \dots, \mathcal{A}_n$	$\mathcal{A}_1, \dots, \mathcal{A}_n$	Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \dots, \mathcal{A}_n$
kohlhase-omdoc_F00187	150	■ 0.993	$\mathcal{C}_1, \dots, \mathcal{C}_m$	$\mathcal{C}_1, \dots, \mathcal{C}_m$	and conclusion elements $\mathcal{C}_1, \dots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$
kohlhase-omdoc_F00188	150	■ 0.993	$\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$	$\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$	and conclusion elements $\mathcal{C}_1, \dots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \dots, \mathcal{A}_n\}$
kohlhase-omdoc_F00189	150	■ 1.000	$\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$	$\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$	and $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assump-
kohlhase-omdoc_F00190	150	■ 1.000	$\mathcal{A}_i$	$\mathcal{A}_i$	tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$, where $\overline{\mathcal{C}}_i$ is a
kohlhase-omdoc_F00191	150	■ 1.000	$\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$	$\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$	tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$, where $\overline{\mathcal{C}}_i$ is a
kohlhase-omdoc_F00192	150	■ 1.000	$\overline{\mathcal{C}}_i$	$\overline{\mathcal{C}}_i$	tions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}}_1, \dots, \overline{\mathcal{C}}_m\}$, where $\overline{\mathcal{C}}_i$ is a
kohlhase-omdoc_F00193	152	■ 0.997	$\mathcal{E}$	$\mathcal{E}$	objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$,
kohlhase-omdoc_F00194	152	■ 0.997	$(\mathcal{W}, \mathbf{A})$	$(\mathcal{W}, \mathbf{A})$	objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$,
kohlhase-omdoc_F00195	152	■ 0.992	$\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$	$\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$	where $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$ is an n -tuple of mathematical objects and $\mathbf{A}$ is an
kohlhase-omdoc_F00196	152	■ 0.923	$\mathbf{S}$	$\mathbf{S}$	symbol $\mathbf{S}$, then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$.

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kohlhase-omdoc_F00197	152	■ 0.923	$\mathbf{S}\left(\mathcal{W}_{\{1\}}, \ldots, \mathcal{W}_{\{n\}}\right)$	$\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$	symbol $\mathbf{S}$, then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$.
kohlhase-omdoc_F00198	152	■ 0.999	$\mathbf{C}$	$\mathbf{C}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,
kohlhase-omdoc_F00199	152	■ 0.999	$\mathcal{Q} \ \mathbf{D}$	$\mathcal{Q}\mathbf{D}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,
kohlhase-omdoc_F00200	152	■ 0.999	$\mathbf{D}$	$\mathbf{D}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,
kohlhase-omdoc_F00201	152	■ 1.000	$\mathcal{Q}$	$\mathcal{Q}$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of
kohlhase-omdoc_F00202	152	■ 1.000	$\mathcal{Q}_{\{1\}} \mathcal{W}_{\{1\}}, \ldots, \mathcal{Q}_{\{m\}} \mathcal{W}_{\{m\}}$	$\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of
kohlhase-omdoc_F00203	152	■ 1.000	$m \geq n = \#\mathcal{W}$	$m \geq n = \#\mathcal{W}$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1\mathcal{W}_1, \dots, \mathcal{Q}_m\mathcal{W}_m$ of $m \geq n = \#\mathcal{W}$ quantifications of
kohlhase-omdoc_F00204	152	■ 0.999	$\mathcal{Q}_{\{i\}}$	$\mathcal{Q}_i$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-
kohlhase-omdoc_F00205	152	■ 0.999	$\forall$	$\forall$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-
kohlhase-omdoc_F00206	152	■ 0.999	$\exists$	$\exists$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-
kohlhase-omdoc_F00207	152	■ 0.999	$\mathcal{Q}^{\prime}$	$\mathcal{Q}'$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-
kohlhase-omdoc_F00208	152	■ 0.999	$m - n$	$m - n$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-
kohlhase-omdoc_F00209	152	■ 0.985	$\mathbf{D}^{\prime}$	$\mathbf{D}'$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent
kohlhase-omdoc_F00210	152	■ 0.985	$\mathcal{W}_{\{i_j\}}$	$\mathcal{W}_{i_j}$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent
kohlhase-omdoc_F00211	152	■ 0.985	$\mathcal{Q}_{\{i_j\}}$	$\mathcal{Q}_{i_j}$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the $\mathcal{W}_{i_j}$ such that the $\mathcal{Q}_{i_j}$ are absent
kohlhase-omdoc_F00212	152	■ 0.998	$\mathcal{W}_{\{j\}}$	$\mathcal{W}_j$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,
kohlhase-omdoc_F00213	152	■ 0.998	$1 \leq j \leq n$	$1 \leq j \leq n$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,
kohlhase-omdoc_F00214	152	■ 0.998	$\mathcal{E} = (\mathcal{W}, \mathbf{A})$	$\mathcal{E} = (\mathcal{W}, \mathbf{A})$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,
kohlhase-omdoc_F00215	152	■ 1.000	$\mathbf{A} = \mathcal{Q}^{\prime} \mathbf{D}^{\prime}$	$\mathbf{A} = \mathcal{Q}'\mathbf{D}'$	then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$, then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$.
kohlhase-omdoc_F00216	152	■ 1.000	$\mathbf{A} = \neg \mathcal{Q}^{\prime} \mathbf{D}^{\prime}$	$\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$	then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$, then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$.
kohlhase-omdoc_F00217	154	■ 1.000	A^*	A^*	alphabet A as an argument and returns the set A^* of stringss over A , <code>concat</code>
kohlhase-omdoc_F00218	154	■ 0.914	ϵ	ϵ	the empty string ϵ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an
kohlhase-omdoc_F00219	154	■ 0.914	$\mathcal{W} = \left(A^*, ::, \epsilon\right)$	$\mathcal{W} = (A^*, ::, \epsilon)$	the empty string ϵ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an
kohlhase-omdoc_F00220	154	■ 1.000	$\left(\mathcal{W}, \mathbf{A}^{\prime}\right)$	$(\mathcal{W}, \mathbf{A}')$	<code>mon.ex2</code> uses the pair $(\mathcal{W}, \mathbf{A}')$ as a counter-example to the false conjecture
kohlhase-omdoc_F00221	154	■ 0.380	$M,$	$M,$	<CMP>In a monoid M , the left unit and the right unit coincide,</CMP>
kohlhase-omdoc_F00223	161	■ 1.000	$\emptyset$	$\emptyset$	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained
kohlhase-omdoc_F00224	161	■ 1.000	ι	ι	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained
kohlhase-omdoc_F00225	161	■ 1.000	$\{a\}$	$\{a\}$	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained

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kohlhase-omdoc_F00226	161	■ 1.000	$\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$	$\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$	e.g. $\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$. This kind of sort is called generated , since it only
kohlhase-omdoc_F00227	162	■ 1.000	$\mathbb{P}$	$\mathbb{P}$	encoding. In this example, we introduce a second sort $\mathbb{P}$ for positive nat-
kohlhase-omdoc_F00228	166	■ 1.000	P	P	Proof: We need to prove that the set P of all prime numbers is not
kohlhase-omdoc_F00229	166	■ 1.000	$P = \{p_1, \dots, p_n\}$	$P = \{p_1, \dots, p_n\}$	3. Then $P = \{p_1, \dots, p_n\}$ for some p_i .
kohlhase-omdoc_F00230	166	■ 1.000	p_i	p_i	3. Then $P = \{p_1, \dots, p_n\}$ for some p_i .
kohlhase-omdoc_F00231	166	■ 1.000	$q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$	$q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$	4. Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$.
kohlhase-omdoc_F00232	166	■ 1.000	$p_i \in P$	$p_i \in P$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.
kohlhase-omdoc_F00233	166	■ 1.000	$q > p_i$	$q > p_i$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.
kohlhase-omdoc_F00234	166	■ 1.000	$q \notin P$	$q \notin P$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.
kohlhase-omdoc_F00235	166	■ 1.000	q	q	6. We prove the absurdity by showing that q is prime:
kohlhase-omdoc_F00236	166	■ 1.000	$q = p_i k + 1$	$q = p_i k + 1$	7. For each $p_i \in P$ we have $q = p_i k + 1$ for some natural
kohlhase-omdoc_F00237	166	■ 1.000	k	k	number k , so p_i can not divide q ;
kohlhase-omdoc_F00238	166	■ 0.990	$\bar{\lambda}\mu\tilde{\mu}$	$\bar{\lambda}\mu\tilde{\mu}$	by interpreting it as an implementation of the $\bar{\lambda}\mu\tilde{\mu}$ calculus [Coe05] proof
kohlhase-omdoc_F00239	168	■ 1.000	$12ff$	$12ff$	the proof. See for example lines $12ff$ in Listing 17.2 for examples of
kohlhase-omdoc_F00240	170	■ 0.986	$[\text{BCF}^+97]$	$[\text{BCF}^+97]$	multiple levels of abstraction). This proof representation (see [BCF ⁺ 97] for a
kohlhase-omdoc_F00241	179	■ 0.791	$(R, +)$	$(R, +)$	R together with two operations $+$ and $*$, such that $(R, +)$ is a group with
kohlhase-omdoc_F00242	179	■ 1.000	$(R^*, *)$	$(R^*, *)$	unit 0 and inverse operation $-$ and $(R^*, *)$ is a monoid with unit 1 and base
kohlhase-omdoc_F00243	179	■ 1.000	$R^* := \{r \in R \mid r \neq 0\}$	$R^* := \{r \in R \mid r \neq 0\}$	set $R^* := \{r \in R \mid r \neq 0\}$. Using the literal inheritance regime introduced so
kohlhase-omdoc_F00244	179	■ 0.604	$M = (S, \circ, e)$	$M = (S, \circ, e)$	² On any monoid $M = (S, \circ, e)$, we have the $\cdot^*$ operator, which converts a set
kohlhase-omdoc_F00245	179	■ 1.000	$S \subseteq M$	$S \subseteq M$	$S \subseteq M$ in to $S^* := \{r \in S \mid r \neq e\}$
kohlhase-omdoc_F00246	179	■ 1.000	$S^* := \{r \in S \mid r \neq e\}$	$S^* := \{r \in S \mid r \neq e\}$	$S \subseteq M$ in to $S^* := \{r \in S \mid r \neq e\}$
kohlhase-omdoc_F00247	180	■ 1.000	$\{t_1, \dots, t_n\}$	${}^3t_1, \dots, t_n$	Let $\mathcal{S}$ be a theory with theory-constitutive elements ³ $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$
kohlhase-omdoc_F00248	180	■ 1.000	$\sigma: \mathcal{S} \rightarrow \mathcal{T}$	$\sigma: \mathcal{S} \rightarrow \mathcal{T}$	Let $\mathcal{S}$ be a theory with theory-constitutive elements ³ $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$
kohlhase-omdoc_F00249	180	■ 1.000	$\sigma(t_i)$	$\sigma(t_i)$	constitutive statements $\sigma(t_i)$ from $\mathcal{S}$. For instance, the theory of rings inherits
kohlhase-omdoc_F00250	180	■ 1.000	$\forall x. x + 0 = x$	$\forall x. x + 0 = x$	the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$
kohlhase-omdoc_F00251	180	■ 1.000	$\sigma(\forall x. x + 0 = x) =$	$\sigma(\forall x. x + 0 = x) =$	the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$
kohlhase-omdoc_F00252	180	■ 0.999	$\forall x. x * 1 = x$	$\forall x. x * 1 = x$	$\forall x. x * 1 = x$.
kohlhase-omdoc_F00253	181	■ 0.999	$R, +, 0, -, *, 1$	$R, +, 0, -, *, 1$	the morphism. In our example, the symbols $R, +, 0, -, *, 1$ are visible in the

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kohlhase-omdoc_ F00254	181	■ 1.000	L	L	concept of a sorting function, i.e. a function that — given a list L as input
kohlhase-omdoc_ F00255	181	■ 1.000	$L^{\{\prime\}}$	L'	— returns a returns a permutation L' of L that is ordered. In the situation
kohlhase-omdoc_ F00256	182	■ 0.879	$\sigma_{\{i\}}$	σ_i	σ_i with <code>xml:id="«refi»"</code> , such that the target theory of σ_{i-1} is the source
kohlhase-omdoc_ F00257	182	■ 0.879	$\sigma_{\{i-1\}}$	σ_{i-1}	σ_i with <code>xml:id="«refi»"</code> , such that the target theory of σ_{i-1} is the source
kohlhase-omdoc_ F00258	182	■ 0.902	$\mathcal{S}, \sigma(\mathbf{S})$	$\mathcal{S}, \sigma(\mathbf{S})$	constitutive statement $\mathbf{S}$ of $\mathcal{S}$, $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a
kohlhase-omdoc_ F00259	182	■ 0.902	$\sigma(\mathbf{S})$	$\sigma(\mathbf{S})$	constitutive statement $\mathbf{S}$ of $\mathcal{S}$, $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a
kohlhase-omdoc_ F00260	182	■ 0.905	$\mathbf{T}$	$\mathbf{T}$	in theory $\mathcal{T}$ via σ , if there is an assertional statement $\mathbf{T}$ in $\mathcal{T}$, such that the
kohlhase-omdoc_ F00261	182	■ 1.000	$\mathcal{T}^5$	$\mathcal{T}^5$	discussed in Section 18.1 induces a theory inclusion from $\mathcal{S}$ into $\mathcal{T}^5$ (the
kohlhase-omdoc_ F00262	183	■ 1.000	$\sigma(\mathbf{A})$	$\sigma(\mathbf{A})$	$\mathcal{S}$ -theorem for for all $\mathcal{T}$ -theorems of the from $\sigma(\mathbf{A})$. If the morphism σ is the
kohlhase-omdoc_ F00263	183	■ 0.421	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	<code><assertion xml:id="conv.assoc">$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$</assertion></code>
kohlhase-omdoc_ F00264	183	■ 0.421	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	<code><assertion xml:id="conv.assoc">$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$</assertion></code>
kohlhase-omdoc_ F00265	183	■ 0.765	$\forall x, y \in M. y \circ x \in M$	$\forall x, y \in M. y \circ x \in M$	<code><assertion xml:id="conv.closed" theory="semigroup">$\forall x, y \in M. y \circ x \in M$</assertion></code>
kohlhase-omdoc_ F00266	183	■ 0.325	$\forall x \in M. e \circ x = x$	$\forall x \in M. e \circ x = x$	<code><assertion xml:id="left.unit" theory="monoid">$\forall x \in M. e \circ x = x$</assertion></code>
kohlhase-omdoc_ F00267	186	■ 0.998	BCD^{+02}	BCD^{+02}	(see e.g. BCD^{+02} for an overview over the branch cut situation). Reading
kohlhase-omdoc_ F00268	187	■ 1.000	Γ	Γ	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00269	187	■ 1.000	$\mathcal{T}_1$	$\mathcal{T}_1$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00270	187	■ 1.000	$\mathcal{T}_2$	$\mathcal{T}_2$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00271	187	■ 1.000	$\mathcal{C}_1$	$\mathcal{C}_1$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00272	187	■ 0.999	$\mathcal{A}_1$	$\mathcal{A}_1$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00273	187	■ 0.999	$\mathcal{B}_1$	$\mathcal{B}_1$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00274	187	■ 0.999	$\mathcal{C}_2$	$\mathcal{C}_2$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00275	187	■ 0.999	$\mathcal{A}_2$	$\mathcal{A}_2$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00276	187	■ 1.000	$\mathcal{B}_2$	$\mathcal{B}_2$	and $\mathcal{B}_2$ (in the latter, we have a non-trivial non-trivial morphism σ). Let us
kohlhase-omdoc_ F00277	187	■ 1.000	$\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$	$\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_ F00278	187	■ 1.000	$\mathcal{X}_1$	$\mathcal{X}_1$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_ F00279	187	■ 1.000	$\mathcal{X}_2$	$\mathcal{X}_2$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_ F00280	187	■ 0.995	α, β	α, β	arrows in Figure 18.8 and have the formula-mappings α , β , and γ) are easy
kohlhase-omdoc_ F00281	187	■ 0.995	γ	γ	arrows in Figure 18.8 and have the formula-mappings α , β , and γ) are easy

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kohlhase-omdoc_ F00282	187	■ 1.000	<code>\beta(\mathbf{B})</code>	$\beta(\mathbf{B})$	constitutive for $\mathcal{C}_1$, say that it is local in $\mathcal{B}_1$, then we already know that $\beta(\mathbf{B})$ is
kohlhase-omdoc_ F00283	187	■ 1.000	<code>\beta</code>	β	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_ F00284	187	■ 1.000	<code>\sigma(\beta(\mathbf{B}))</code>	$\sigma(\beta(\mathbf{B}))$	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_ F00285	187	■ 1.000	<code>\Gamma=\sigma \circ \beta</code>	$\Gamma = \sigma \circ \beta$	have $\Gamma = \sigma \circ \beta$, then we are done for $\mathbf{B}$ and in fact for all local statements
kohlhase-omdoc_ F00286	187	■ 1.000	<code>\Gamma^{-1}</code>	Γ^{-1}	(and often also in the converse theory inclusion Γ^{-1} that would prove that the
kohlhase-omdoc_ F00287	188	■ 1.000	<code>\tau:=\sigma_{\{n\}} \circ \cdots \circ \sigma_1 \circ \sigma</code> <code>\sigma</code>	$\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$	morphism $\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local
kohlhase-omdoc_ F00288	188	■ 1.000	<code>\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1</code>	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$	morphism $\tau := \sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local
kohlhase-omdoc_ F00289	188	■ 1.000	<code>\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}</code>	$\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$	theory import) and $\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$ are theory inclusions (or local theory import).
kohlhase-omdoc_ F00290	188	■ 1.000	<code>\mathcal{T}_1, \mathcal{T}_2, \mathcal{A}_1</code>	$\mathcal{T}_1, \mathcal{T}_2, \mathcal{A}_1$	justify four new axiom inclusions from the theories $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{A}_1$, and $\mathcal{B}_1$ into
kohlhase-omdoc_ F00291	188	■ 1.000	<code>\mathcal{X}</code>	$\mathcal{X}$	Thus, for each theory $\mathcal{X}$ that $\mathcal{C}_1$ inherits from, there is an axiom inclusion
kohlhase-omdoc_ F00292	188	■ 1.000	<code>\mathcal{S} \xrightarrow{\sigma} \mathcal{T}</code>	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$	We call a situation where we have a formula mapping $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$, and
kohlhase-omdoc_ F00293	188	■ 1.000	<code>\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}</code>	$\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$	an axiom inclusion $\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$ for every theory $\mathcal{X}$ that $\mathcal{S}$ inherits from a
kohlhase-omdoc_ F00294	188	■ 1.000	<code>\sigma_{\mathcal{X}}</code>	$\sigma_{\mathcal{X}}$	decomposition for σ , if the $\sigma_{\mathcal{X}}$ and σ are compatible. As we have seen in
kohlhase-omdoc_ F00295	190	■ 0.996	<code>i</code>	i	there must be theory inclusions σ_i with <code>xml:id="refi"</code> , such that the target
kohlhase-omdoc_ F00296	190	■ 1.000	<code>\sigma_1</code>	σ_1	theory of σ_1 is the source theory of σ_2 .
kohlhase-omdoc_ F00297	190	■ 1.000	<code>\sigma_2</code>	σ_2	theory of σ_1 is the source theory of σ_2 .
kohlhase-omdoc_ F00298	197	■ 1.000	<code>\forall P, Q. P \vee Q \Rightarrow Q \vee P</code>	$\forall P, Q. P \vee Q \Rightarrow Q \vee P$	$\forall P, Q. P \vee Q \Rightarrow Q \vee P$ assuming appropriate templates for implication and
kohlhase-omdoc_ F00299	199	■ 1.000	<code>g c d</code>	gcd	gcd in English but ggT (“größter gemeinsamer Teiler”) in German). Obvi-
kohlhase-omdoc_ F00300	199	■ 1.000	<code>g g T</code>	ggT	gcd in English but ggT (“größter gemeinsamer Teiler”) in German). Obvi-
kohlhase-omdoc_ F00301	200	■ 1.000	<code>f^{\prime}</code>	f'	f' . The case <code>infix</code> is reserved for binary operators, where the function is
kohlhase-omdoc_ F00306	201	■ 1.000	<code>(a, b)</code>	(a, b)	presentation. For instance, in the notation (a, b) for pairs, the binary
kohlhase-omdoc_ F00307	201	■ 0.912	<code>\mathrm{IAT}_{\mathrm{E}}\mathrm{X}</code>	$\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$	of the figure: the $\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$ and the HTML presentations have attached
kohlhase-omdoc_ F00308	202	■ 0.369	<code>\forall X</code>	$\forall X$	MATH expression yields $\forall X.\text{true}$, where the glyph $\forall$ carries a hyperlink ⁶ to
kohlhase-omdoc_ F00309	202	■ 0.369	<code>{ }^{\varnothing}</code>	$\varnothing$	MATH expression yields $\forall X.\text{true}$, where the glyph $\forall$ carries a hyperlink ⁶ to
kohlhase-omdoc_ F00310	202	■ 1.000	<code>\binom{n}{m}</code>	$\binom{n}{m}$	For instance, the binomial coefficient is some presented as $\binom{n}{m}$ (spoken
kohlhase-omdoc_ F00311	202	■ 1.000	<code>m</code>	m	“ n choose m ”) and represented as
kohlhase-omdoc_ F00312	204	■ 0.995	<code>X_{\{4\}}</code>	X_4	tifier would be represented as X_4 , yielding $\forall X_4. X_4 = X_4$ which is exactly

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kohlhase-omdoc_ F00313	204	■ 0.995	$\forall x_4. x_4 = x_4$	$\forall X_4. X_4 = X_4$	tifier would be represented as X_4 , yielding $\forall X_4. X_4 = X_4$ which is exactly
kohlhase-omdoc_ F00314	205	■ 0.982	$\forall \prec, X. X \prec X$	$\forall \prec, X. X \prec X$	This would give us the presentation $\forall \prec, X. X \prec X$. Here, the first occurrence
kohlhase-omdoc_ F00315	205	■ 1.000	$\prec$	$\prec$	of the variable $\prec$ is handled by the second notation declaration (it does not
kohlhase-omdoc_ F00317	215	■ 0.891	$\{\backslash, p \mid \hbox{p and $p+2$ are prime}\backslash, \}$	<i>(not rendered)</i>	$\{\backslash, p \mid \hbox{p and $p+2$ are prime}\backslash, \}$
kohlhase-omdoc_ F00318	215	■ 0.891	and	<i>and</i>	$\{\backslash, p \mid \hbox{p and $p+2$ are prime}\backslash, \}$
kohlhase-omdoc_ F00319	215	■ 0.891	are prime $\backslash, \}$	<i>(not rendered)</i>	$\{\backslash, p \mid \hbox{p and $p+2$ are prime}\backslash, \}$
kohlhase-omdoc_ F00320	215	■ 0.468	$\{\backslash, p \mid p$	$\{p \mid p$	$\{\backslash, p \mid p \text{~and~} p+2 \text{ are prime} \backslash, \}$
kohlhase-omdoc_ F00321	215	■ 0.468	$p+2$	$p + 2$	$\{\backslash, p \mid p \text{~and~} p+2 \text{ are prime} \backslash, \}$
kohlhase-omdoc_ F00322	215	■ 0.468	$\backslash, \}$		$\{\backslash, p \mid p \text{~and~} p+2 \text{ are prime} \backslash, \}$
kohlhase-omdoc_ F00323	215	■ 0.405	$\{\backslash, p \mid p \backslash \{\rm and\} p+2 \rm \ are \ prime \backslash, \}$	<i>(not rendered)</i>	$\{\backslash, p \mid p \backslash \{\rm and\} \ p+2 \rm \ are \ prime \backslash, \}$ </phrase>
kohlhase-omdoc_ F00324	218	■ 1.000	$\mathcal{K}$	$\mathcal{K}$	For a formal definition let $\mathcal{K}$ be a set of documents. We take a docu-
kohlhase-omdoc_ F00325	218	■ 0.935	$\{ \}^{\{1\}} \mathcal{X}$	$^1 \mathcal{X}$	ment model to be a partial equivalence relation ¹ $\mathcal{X}$ on documents, such
kohlhase-omdoc_ F00326	218	■ 0.998	$\{d \mid d \in \mathcal{X}\} = \mathcal{K}$	$\{d \mid d \in \mathcal{X}\} = \mathcal{K}$	that $\{d \mid d \in \mathcal{X}\} = \mathcal{K}$. In particular, a relation $\mathcal{X}$ is an equivalence relation on
kohlhase-omdoc_ F00327	218	■ 1.000	d	d	$\mathcal{K}$. For a given document model $\mathcal{X}$, let us say that two documents d and d'
kohlhase-omdoc_ F00328	218	■ 1.000	$d^{\{\prime\}}$	d'	$\mathcal{K}$. For a given document model $\mathcal{X}$, let us say that two documents d and d'
kohlhase-omdoc_ F00329	218	■ 1.000	$d \in \mathcal{X}, d^{\{\prime\}}$	$d \mathcal{X} d'$	are $\mathcal{X}$-equal , iff $d \mathcal{X} d'$. We call a property p $\mathcal{X}$-invariant , iff for all $d \mathcal{X} d', p$
kohlhase-omdoc_ F00330	218	■ 1.000	$p \in \mathcal{X}$	$p \mathcal{X}$	are $\mathcal{X}$-equal , iff $d \mathcal{X} d'$. We call a property p $\mathcal{X}$-invariant , iff for all $d \mathcal{X} d', p$
kohlhase-omdoc_ F00331	218	■ 1.000	$d \in \mathcal{X}, d^{\{\prime\}}, p$	$d \mathcal{X} d', p$	are $\mathcal{X}$-equal , iff $d \mathcal{X} d'$. We call a property p $\mathcal{X}$-invariant , iff for all $d \mathcal{X} d', p$
kohlhase-omdoc_ F00332	218	■ 1.000	p	p	holds on d whenever p holds on d' .
kohlhase-omdoc_ F00333	218	■ 1.000	$[d]_{\mathcal{X}}$	$[d]_{\mathcal{X}}$	¹ A partial equivalence relation is a symmetric transitive relation. We will use $[d]_{\mathcal{X}}$
kohlhase-omdoc_ F00334	218	■ 0.999	$[d]_{\mathcal{X}} := \{e \mid e \in \mathcal{X}\}$	$[d]_{\mathcal{X}} := \{e \mid e \in \mathcal{X}\}$	for the equivalence class of d , i.e. $[d]_{\mathcal{X}} := \{e \mid e \in \mathcal{X}\}$
kohlhase-omdoc_ F00335	220	■ 1.000	$X+Y=$	$X + Y =$	<i>not</i> use mathematical equality here, which would make the formula $X + Y =$
kohlhase-omdoc_ F00336	220	■ 0.268	$Y+X$	$Y + X$	$Y + X$ (the <code>om:OMOBJ</code> with <code>xml:id="comm1"</code> in Listing 22.3 instantiated with
kohlhase-omdoc_ F00337	220	■ 1.000	$X+Y=X+Y$	$X + Y = X + Y$	addition for <code>op</code>) mathematically equal to the trivial condition $X+Y = X+Y$,
kohlhase-omdoc_ F00338	220	■ 1.000	$X+Y$	$X + Y$	obtained by exchanging the right hand side $Y + X$ of the equality by $X + Y$,
kohlhase-omdoc_ F00339	232	■ 0.999	$\left[\mathrm{ABD}^{+01}\right]$	$[\mathrm{ABD}^{+01}]$	tions for DTD modularization $[\mathrm{ABD}^{+01}]$. This partitions the DTD into DTD
kohlhase-omdoc_ F00340	232	■ 0.966	$\{ \}^{\{+\}} 01$	$^{+01}$	process, we will briefly describe the module structure. Following $[\mathrm{ABD}^{+01}]$,
kohlhase-omdoc_ F00341	234	■ 0.988	$d \rangle$	$d \rangle$	as <code>omdoc-$\langle \langle \mathrm{SIId} \rangle \rangle$.rnc</code> , where $\langle \langle \mathrm{SIId} \rangle \rangle$ stands for the sub-language identifier.

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kohlhase-omdoc_ F00342	238	■ 1.000	$2\ n$	$2n$	so achieve interoperability for n systems with only $2n$ translations instead of
kohlhase-omdoc_ F00343	238	■ 1.000	$n^{\{2\}}$	n^2	n^2 . As we have seen in Section 2.1, OPENMATH and Content-MATHML pro-
kohlhase-omdoc_ F00344	240	■ 0.970	$\lambda\mathrm{C\ l\ a\ m}$	$\lambda Clam$	systems Ω MEGA [BCF ⁺ 97], INKA [HS96], Pvs [ORS92], $\lambda Clam$ [RSG98b],
kohlhase-omdoc_ F00345	240	■ 0.836	$\left[\mathrm{ABI}^{\{+\}}96\right]$	$[\mathrm{ABI}^+96]$	TPS [ABI⁺96] and CoQ [Tea] see Section 26.18
kohlhase-omdoc_ F00346	241	■ 0.183	$\mathrm{IAT}_{\{E\}}\ \mathrm{X}$	$\mathrm{IAT}_E\mathrm{X}$	omdoc2html.xsl for XHTML+MATHML and omdoc2tex.xsl for L ^A T _E X.
kohlhase-omdoc_ F00347	243	■ 0.990	$\mathrm{T}_{\{\mathrm{E}\}}\ \mathrm{X}_{\{\mathrm{MACS}\}}$	$\mathrm{T}_E\mathrm{X}_{\mathrm{MACS}}$	plugin for MOZILLA-based browsers, and the plugin for T _E X _{MACS} (Sec-
kohlhase-omdoc_ F00348	243	■ 1.000	$\mathrm{ST}_{\{\mathrm{E}\}}\ \mathrm{X}$	$\mathrm{ST}_E\mathrm{X}$	OMDOC that can be generated by the qmath parser. sT _E X defines a
kohlhase-omdoc_ F00349	243	■ 0.220	$_{\{\mathrm{E}\}}$	E	XHTML+MATHML or L ^A T _E X, which can in turn be published on the
kohlhase-omdoc_ F00350	246	■ 1.000	$2+2=4$	$2 + 2 = 4$	Freedom is the freedom to say $\$2+2=4\$$.
kohlhase-omdoc_ F00351	251	■ 0.429	$\$ \ \mathrm{e}^{\wedge\{\mathrm{pi}\}} + 1 = 0 \ \$$	$\$e^{\wedge(\mathrm{pi} * i)} + 1 = 0\$$	is similar to typing T _E X or QMATH text: one can type $\$e^{\wedge(\mathrm{pi} * i)} + 1 = 0\$$ and
kohlhase-omdoc_ F00352	251	■ 1.000	$e^{\wedge\{\mathrm{pi}\ i\}} + 1 = 0$	$e^{\pi i} + 1 = 0$	get $e^{\pi i} + 1 = 0$ without having to look at the formula editor or use the
kohlhase-omdoc_ F00353	251	■ 0.370	$\{\}^{\{\mathrm{TM}\}}$	TM	“%pi” for MAXIMA, or “Pi” for YACAS, MAPLE [™] and MATHEMATICA [®] .
kohlhase-omdoc_ F00354	251	■ 0.370	$\{\}^{\{\mathrm{R}\}}$	(R)	“%pi” for MAXIMA, or “Pi” for YACAS, MAPLE [™] and MATHEMATICA [®] .
kohlhase-omdoc_ F00355	254	■ 1.000	0	O	servers (see [KF01] for details). In particular, an object O residing on an
kohlhase-omdoc_ F00356	254	■ 1.000	$0^{\{\mathrm{prime}\}}$	O'	MBASE server S can refer to (or depend on) an object O' residing on a
kohlhase-omdoc_ F00357	254	■ 1.000	$S^{\{\mathrm{prime}\}}$	S'	server S' ; a query to O that needs information about O' will be delegated to
kohlhase-omdoc_ F00358	255	■ 1.000	$\mathrm{binom}\{n\}\{k\}, \{\}_n\ C^{\{k\}}, C_{\{k\}}^{\{n\}}$	$\binom{n}{k}, {}_nC^k, C_k^n$	$\binom{n}{k}$, ${}_nC^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_ F00359	255	■ 1.000	$C_{\{n\}}^{\{k\}}$	C_n^k	$\binom{n}{k}$, ${}_nC^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_ F00360	255	■ 1.000	$\frac{n!}{k!(n-k)!}$	$\frac{n!}{k!(n-k)!}$	$\binom{n}{k}$, ${}_nC^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_ F00361	255	■ 0.999	$\int f(x)\ d\ x$	$\int f(x)dx$	jects, e.g. an integral expression of the form $\int f(x)dx$ can mean a Rie-
kohlhase-omdoc_ F00362	255	■ 1.000	$\int f(y)\ d\ y$	$\int f(y)dy$	stance $\int f(x)dx$ means the same as $\int f(y)dy$ (modulo α -equivalence), so
kohlhase-omdoc_ F00363	255	■ 1.000	$s(t)$	$s(t)$	cal problem such as finding the power of a given signal $s(t)$. Of course our
kohlhase-omdoc_ F00364	255	■ 1.000	$\int_{\{?\}}^{\{?\}} s^{\{2\}}(t)\ d\ t$	$\int_{\{?\}}^{\{?\}} s^2(t)dt$	MATHWEBSEARCH to search for something that looks like $\int_{\{?\}}^{\{?\}} s^2(t)dt$ (for
kohlhase-omdoc_ F00365	255	■ 1.000	$\frac{1}{T} \int_{\{0\}}^{\{T\}} s^{\{2\}}(t)\ d\ t = \Sigma_{k=-\infty}^{\infty} c_{\{k\}} ^2$	$\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty} c_k ^2$	$\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty} c_k ^2$ where c_k are the Fourier coefficients of the
kohlhase-omdoc_ F00366	255	■ 1.000	$c_{\{k\}}$	c_k	$\frac{1}{T} \int_0^T s^2(t)dt = \Sigma_{k=-\infty}^{\infty} c_k ^2$ where c_k are the Fourier coefficients of the
kohlhase-omdoc_ F00367	257	■ 1.000	$A=B$	$A = B$	While of searching expressions of the form $A = B$, we might like to find

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kohlhase-omdoc_ F00368	257	■ 0.737	B=A	$B = A$	occurrences of $B = A$ as well. At this point the <code>mq:anyorder</code> attribute comes
kohlhase-omdoc_ F00369	258	■ 0.994	$\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty} c_k ^2$	$\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty} c_k ^2$	the variant $\frac{1}{T} \int_0^T s^2(x) dx = \Sigma_{k=-\infty}^{\infty} c_k ^2$.
kohlhase-omdoc_ F00370	259	■ 0.319	$\left. \{ \}^{+} \right\} \text{03}, \text{KBL} \{ \}^{+} \text{04}, \text{KBKB}^{+} \text{04} \right]$	$^{+03}, \text{KBL}^{+04}, \text{KBKB}^{+04}]$	(see [KBHL ⁺⁰³ , KBLL ⁺⁰⁴ , KBKB ⁺⁰⁴]).
kohlhase-omdoc_ F00371	261	■ 1.000	T	T	entities, such that (for a theorem T a proof P , and sections A, B):
kohlhase-omdoc_ F00372	261	■ 1.000	A, B	A, B	entities, such that (for a theorem T a proof P , and sections A, B):
kohlhase-omdoc_ F00373	261	■ 0.533	P & A	$P \& A$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_ F00374	261	■ 0.533	T & P	$T \& P$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_ F00375	261	■ 0.533	T $\Rightarrow$ B	$T \Rightarrow B$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_ F00376	264	■ 0.997	$\mathbb{F}$	$\mathbb{F}$	ematical objects ‘live’. For instance, when a chapter begins with “Let $\mathbb{F}$ be
kohlhase-omdoc_ F00377	264	■ 1.000	$\frac{3}{4} / \frac{2}{3} / \frac{3}{5}$	$\frac{3/4 + 2/3}{5}$	fraction display. In $\frac{3/4+2/3}{5}$ we have used them both.
kohlhase-omdoc_ F00378	264	■ 0.998	$3, 4 \in \mathbb{Z}$	$3, 4 \in \mathbb{Z}$	$3, 4 \in \mathbb{Z}$. The corresponding OPENMATH expression would be the equivalent
kohlhase-omdoc_ F00379	264	■ 1.000	$3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$	$3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$	of $\mathbb{Z} \wedge 4 \in \mathbb{Z}$, but our OPENMATH statement reads that the sequence
kohlhase-omdoc_ F00380	264	■ 0.998	$\mathbb{Z}$	$\mathbb{Z}$	$3, 4$ belongs to $\mathbb{Z}$. So here, the semantics of the element-of symbol has been
kohlhase-omdoc_ F00381	268	■ 0.608	$\left[\text{MGH}^{+} \right] \text{05}$	[MGH ⁺⁰⁵]	and tools [MGH ⁺⁰⁵].
kohlhase-omdoc_ F00382	269	■ 0.995	$\left[\text{KAB}^{+} \right] \text{04}$	[KAB ⁺⁰⁴]	ment (PISA) [KAB ⁺⁰⁴] and National Council of Teachers of Mathematics
kohlhase-omdoc_ F00383	273	■ 0.996	$10^{\prime}000$	10’000	To date, more than 10’000 <i>items</i> of OMDoc content has been written
kohlhase-omdoc_ F00384	273	■ 1.000	$1, \ldots, k, \ldots, n$	$1, \dots, k, \dots, n$	age of the ellipsis in such a formula as $\underline{1, \dots, k, \dots, n}$ is a simple example of
kohlhase-omdoc_ F00385	275	■ 0.998	$\left[\text{SBB}^{+} \right] \text{06}$	[SBB ⁺⁰⁶]	learning [SBB ⁺⁰⁶].
kohlhase-omdoc_ F00386	279	■ 1.000	$\{ \}^{\sim}$	$\sim$	http://www.cs.nott.ac.uk/~lad/research/
kohlhase-omdoc_ F00388	282	■ 0.757	$\{ \}^{+} \text{00}$	$^{+00}$	SL [AHL ⁺⁰⁰]. All these specification languages provide constructs similar
kohlhase-omdoc_ F00389	286	■ 0.830	$\left[\text{SHB}^{+} \right] \text{99}$	[SHB ⁺⁹⁹]	of the GUI, shared with the Ω MEGA-system [SHB ⁺⁹⁹], are written
kohlhase-omdoc_ F00390	295	■ 0.465	$\text{St}_{\text{E}} \{ \text{E} \} \text{X}$	St _E X	We present $\mathfrak{S}\text{T}_{\text{E}}\text{X}$ (Semantic T _E X) a collection of macro packages for
kohlhase-omdoc_ F00391	295	■ 0.962	$\text{ST}_{\text{E}} \{ \text{E} \}^{\sim} \{ \text{X} \}$	ST _E ^X	documents can be authored and maintained in $\mathfrak{S}\text{T}_{\text{E}}\text{X}$ using a simple text ed-
kohlhase-omdoc_ F00392	295	■ 0.580	$\text{T}_{\text{E}} \{ \text{E} \} \text{X} / \text{IAT}_{\text{E}} \{ \text{E} \} \text{X}$	T _E X/IAT _E X	– The T _E X/L ^A T _E X syntax is much more compact than OMDoc, and if
kohlhase-omdoc_ F00393	297	■ 0.713	$\binom{n}{k}, \{ \}_n C^k, \mathcal{C}_{\{k\}}^n$	$\binom{n}{k}, {}_n C^k, \mathcal{C}_k^n$	tations” ¹⁹ for binomial coefficients: $\binom{n}{k}$, ${}_n C^k$, $\mathcal{C}_k^n$, and $\mathcal{C}_n^k$ all mean the same
kohlhase-omdoc_ F00394	297	■ 0.713	$\mathcal{C}_{\{n\}}^k$	$\mathcal{C}_n^k$	tations” ¹⁹ for binomial coefficients: $\binom{n}{k}$, ${}_n C^k$, $\mathcal{C}_k^n$, and $\mathcal{C}_n^k$ all mean the same
kohlhase-omdoc_ F00395	297	■ 1.000	$\sum_{k=1}^{\infty} x^k$	$\sum_{k=1}^{\infty} x^k$	for the mathematical expression $\sum_{k=1}^{\infty} x^k$. In the first form, we specify that we

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00396	297	■ 0.943	∞	∞	(i) the bound variable k (ii) the number 1 (iii) ∞ (iv) $\backslash\mathrm{Cexp}\{x\}k$ (i.e. x to
kohlhase-omdoc_F00397	297	■ 0.870	$k=1$	$k = 1$	capital sigma character (Σ) whose subscript is the equation $k=$ and whose
kohlhase-omdoc_F00398	298	■ 0.996	$\sum_{i=1}^n x^i$	$\sum_{i=1}^n x^i$	an expression like $\sum_{i=1}^n x^i$ as $\backslash\mathrm{Sumfromto}\{i\}1n\{2i-1\}$. In this example we
kohlhase-omdoc_F00399	298	■ 1.000	$\backslash\mathrm{arbitrary}n$	(not rendered)	What is the sum of the first $\backslash\mathrm{arbitrary}n$ odd numbers, i.e.
kohlhase-omdoc_F00400	298	■ 1.000	$\backslash\mathrm{Sumfromto}\{i\}1\backslash\mathrm{arbitrary}n\{2i-1\}?$	(not rendered)	$\backslash\mathrm{Sumfromto}\{i\}1\backslash\mathrm{arbitrary}n\{2i-1\}?\$$
kohlhase-omdoc_F00401	298	■ 0.998	$\sum_{i=1}^n 2i - 1$	$\sum_{i=1}^n 2i - 1$	<u>What is the sum of the first n odd numbers, i.e. $\sum_{i=1}^n 2i - 1$?</u>
kohlhase-omdoc_F00402	298	■ 0.660	$\mathrm{S}_{\mathrm{E}}\mathrm{X}$	$\mathrm{S}_{\mathrm{E}}\mathrm{X}$	The $\mathrm{S}_{\mathrm{E}}\mathrm{X}$ macro packages have been validated together with a case
kohlhase-omdoc_F00403	303	■ 0.999	ax^2+bx+c	$ax^2 + bx + c$	mial expression ax^2+bx+c whose internal representation in MATHEMATICA [®]
kohlhase-omdoc_F00405	306	■ 1.000	ρ_*	ρ_*	inclusions ρ_* that interpret the system concepts in terms of <code>mbase:sets</code> .
kohlhase-omdoc_F00406	306	■ 1.000	ρ_*^{-1}	ρ_*^{-1}	the integration theory, using the partial inverse ρ_*^{-1} of ρ_* . For an integration
kohlhase-omdoc_F00407	308	■ 1.000	$\mathcal{S}=(\mathcal{L},\mathcal{C})$	$\mathcal{S} = (\mathcal{L}, \mathcal{C})$	Definition 26.18.1. A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$
kohlhase-omdoc_F00408	308	■ 1.000	$\mathcal{L}$	$\mathcal{L}$	Definition 26.18.1. A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$
kohlhase-omdoc_F00409	308	■ 0.999	$\mathcal{H}$	$\mathcal{H}$	rules). A calculus gives us a notion of a $\mathcal{C}$ -derivation of $\mathbf{A}$ from $\mathcal{H}$, which
kohlhase-omdoc_F00410	309	■ 1.000	$\mathcal{D}:\mathcal{H}\vdash_{\mathcal{C}}\mathbf{A}$	$\mathcal{D}:\mathcal{H}\vdash_{\mathcal{C}}\mathbf{A}$	we will denote by $\mathcal{D}:\mathcal{H}\vdash_{\mathcal{C}}\mathbf{A}$. Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>
kohlhase-omdoc_F00411	309	■ 1.000	$\mathcal{S}'$	$\mathcal{S}'$	we will denote by $\mathcal{D}:\mathcal{H}\vdash_{\mathcal{C}}\mathbf{A}$. Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>
kohlhase-omdoc_F00412	309	■ 1.000	$\mathcal{F}:\mathcal{S}\rightarrow\mathcal{S}'$	$\mathcal{F}:\mathcal{S}\rightarrow\mathcal{S}'$	<i>morphism</i> $\mathcal{F}:\mathcal{S}\rightarrow\mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}}:\mathcal{L}\rightarrow\mathcal{L}'$ and
kohlhase-omdoc_F00413	309	■ 1.000	$\mathcal{F}^{\mathcal{L}}:\mathcal{L}\rightarrow\mathcal{L}'$	$\mathcal{F}^{\mathcal{L}}:\mathcal{L}\rightarrow\mathcal{L}'$	<i>morphism</i> $\mathcal{F}:\mathcal{S}\rightarrow\mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}}:\mathcal{L}\rightarrow\mathcal{L}'$ and
kohlhase-omdoc_F00414	309	■ 1.000	$\mathcal{F}^{\mathcal{D}}$	$\mathcal{F}^{\mathcal{D}}$	a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that
kohlhase-omdoc_F00415	309	■ 1.000	$\mathcal{C}'$	$\mathcal{C}'$	a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that
kohlhase-omdoc_F00416	309	■ 0.999	$\mathcal{F}^{\mathcal{D}}(\mathcal{D}):\mathcal{F}^{\mathcal{L}}(\mathcal{H})\vdash_{\mathcal{C}'}\mathcal{F}^{\mathcal{L}}(\mathbf{A})$	$\mathcal{F}^{\mathcal{D}}(\mathcal{D}):\mathcal{F}^{\mathcal{L}}(\mathcal{H})\vdash_{\mathcal{C}'}\mathcal{F}^{\mathcal{L}}(\mathbf{A})$	for any $\mathcal{C}$ -derivation $\mathcal{D}:\mathcal{H}\vdash_{\mathcal{C}}\mathbf{A}$ we have $\mathcal{F}^{\mathcal{D}}(\mathcal{D}):\mathcal{F}^{\mathcal{L}}(\mathcal{H})\vdash_{\mathcal{C}'}\mathcal{F}^{\mathcal{L}}(\mathbf{A})$.
kohlhase-omdoc_F00417	309	■ 1.000	$\mathcal{R}$	$\mathcal{R}$	instance the morphism $\mathcal{R}$ from sorted first-order logic ($\mathcal{S}FOL$) to unsorted
kohlhase-omdoc_F00418	309	■ 1.000	$(\mathcal{S}FOL)$	$(\mathcal{S}FOL)$	instance the morphism $\mathcal{R}$ from sorted first-order logic ($\mathcal{S}FOL$) to unsorted
kohlhase-omdoc_F00419	309	■ 1.000	(FOL)	(FOL)	first-order logic (FOL). For every sorted constant $\mathcal{R}$ introduces an axiom
kohlhase-omdoc_F00420	309	■ 1.000	$\mathcal{R}([+:\mathbb{N}\rightarrow\mathbb{N}\rightarrow\mathbb{N}])=\forall X,Y.\mathbb{N}(X)\wedge\mathbb{N}(Y)\Rightarrow\mathbb{N}(X+Y)$	$\mathcal{R}([+:\mathbb{N}\rightarrow\mathbb{N}\rightarrow\mathbb{N}])=\forall X,Y.\mathbb{N}(X)\wedge\mathbb{N}(Y)\Rightarrow\mathbb{N}(X+Y)$	e.g. $\mathcal{R}([+:\mathbb{N}\rightarrow\mathbb{N}\rightarrow\mathbb{N}])=\forall X,Y.\mathbb{N}(X)\wedge\mathbb{N}(Y)\Rightarrow\mathbb{N}(X+Y)$. On formu-
kohlhase-omdoc_F00421	309	■ 0.999	$\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A})=\forall X.\mathbb{B}(X)\Rightarrow\mathcal{R}(\mathbf{A})$	$\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A})=\forall X.\mathbb{B}(X)\Rightarrow\mathcal{R}(\mathbf{A})$	predicates, e.g. $\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A})=\forall X.\mathbb{B}(X)\Rightarrow\mathcal{R}(\mathbf{A})$. Finally, for proofs we have
kohlhase-omdoc_F00422	309	■ 1.000	$\mathbb{A},\mathbb{B},\dots$	$\mathbb{A},\mathbb{B},\dots$	the correspondence given in Figure 26.26, where $\mathbb{A},\mathbb{B},\dots$ are sort symbols.

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kohlhase-omdoc_F00454	384	■ 0.952	$\mathcal{U}$	$\mathcal{U}$	<i>Lovely ΩMEGA User Interface. Formal Aspects of Computing, 11:326–342, 1999.</i>
kohlhase-omdoc_F00455	384	■ 0.952	$\mathcal{I}$	$\mathcal{I}$	<i>Lovely ΩMEGA User Interface. Formal Aspects of Computing, 11:326–342, 1999.</i>
kohlhase-omdoc_F00456	386	■ 0.604	X I I I, 23,25,239,285,304	$XIII$, 23, 25, 239, 285, 304	Ω MEGA $XIII$, 23, 25, 239, 285, 304,
kohlhase-omdoc_F00457	386	■ 1.000	$\bar{\lambda} \mu E \mu$	$\bar{\lambda} \mu E \mu$	$\bar{\lambda} \mu E \mu$
kohlhase-omdoc_F00458	421	■ 0.942	$\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$	$\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$	$\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\mathrm{MACS}} 309$